



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

NEURAL NETWORKS AND DEEP LEARNING [CSE-3273]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how least-mean-square algorithm minimizes the instantaneous value of the cost function and instantaneous estimate of the weight vector in the neural network. (5)
 - A)
 - B) Compare batch learning and online learning with example. (3)
 - C) Demonstrate the probability of dichotomies belonging to two different classes using Cover's theorem on separability of patterns. (2)
- 2) Apply recursive least-squares estimation for finding the optimal weight vector in the neural network. (5)
 - A)
 - B) Make use of Willshaw von der Malsburg and Kohonen feature mapping models to explain how the neural network can self-organize to create structured feature maps (3)
 - C) Compare batch mode and stochastic mode of training a multilayer perceptron. (2)
- 3) Identify the three important essential conditions to ensure the stability conditions of the Hopfield Network. (5)
 - A)
 - B) Simplify the following input x using outer product method of associate memories to find the weight matrix. (3)
$$x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$
 - C) Outline the gradient descent algorithm and discuss the various types of gradient descent. (2)
- 4) Make use of MNIST dataset to write a python program to analyze the performance of the model with dropout and without dropout. (5)
 - A)
 - B) Compare Stochastic gradient descent and Momentum. (3)

- C) Analyze the impact of batch normalization on the generalization performance of neural networks. (2)
- 5) Identify the role of filters in Convolutional Neural Network (CNN). Make use of CNN to determine the shape of output matrix of an image of size 15 x 15 that uses a padding size of 2, stride size 2, and a 5 x 5 filter. (5)
- A)
- B) Analyze the use of Restricted Boltzmann Machines (RBMs) for implementing the Recommender system. (3)
- C) Identify four different real time applications of using Recurrent Neural Network (RNN). (2)

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